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Department:
Basic Education
REPUBLIC OF SOUTH AFRICA

NATIONAL SENIOR CERTIFICATE/ NASIONALE SENIOR SERTIFIKAAT

GRADE/GRAAD 12

**MATHEMATICAL LITERACY P2/
WISKUNDIGE GELETTERDHEID V2**

NOVEMBER 2024

MARKING GUIDELINES/NASIENRIGLYNE

MARKS/PUNTE: 150

Symbol/Kode	Explanation/Verduideliking
MA	Method with accuracy/ <i>Metode met akkuraatheid</i>
MCA	Method with consistent accuracy/ <i>Metode met volgehoue akkuraatheid</i>
CA	Consistent accuracy/ <i>Volgehoue akkuraatheid</i>
A	Accuracy/ <i>Akkuraatheid</i>
C	Conversion/ <i>Herleiding</i>
S	Simplification/ <i>Vereenvoudiging</i>
RT	Reading from a table/graph/document/diagram/ <i>Lees vanaf tabel/grafiek/dokument/diagram</i>
SF	Correct substitution in a formula/ <i>Korrekte vervanging in 'n formule</i>
O	Opinion/Explanation/ <i>Opinie/Verduideliking</i>
P	Penalty, e.g. for no units, incorrect rounding off, etc./ <i>Penalising, bv. vir geen eenhede, verkeerde afronding, ens.</i>
NPR	No penalty for correct rounding/ <i>Geen penalising vir korrekte afronding nie</i>
NPU	No penalty for omitting unit, but wrong unit is penalised/ <i>Geen penalising indien die eenheid uitgelos is nie, maar wel indien 'n verkeerde eenheid gebruik word.</i>
AO	Answer only/ <i>Slegs antwoord</i>

**These marking guidelines consist of 17 pages.
*Hierdie nasienriglyne bestaan uit 17 bladsye.***

NOTE:

- If a candidate answers a question TWICE, only mark the FIRST attempt.
- If a candidate has crossed out (cancelled) an attempt to a question and NOT redone the solution, mark the crossed out (cancelled) version.
- Consistent accuracy (CA) applies in ALL aspects of the marking guidelines; however it stops at the second calculation error.
- If the candidate presents any extra solution when reading from a graph, table, layout plan and map, then penalise for every extra item presented.
- Rounding is an independent mark.
- General principle of marking, if the candidate makes one mistake one mark is deducted.
- A conclusion mark can only be given if $\frac{1}{3}$ of the total marks for the sub-question have been awarded.
- No penalty for rounding (NPR) if the first decimal is correct.

LET WEL:

- As 'n kandidaat 'n vraag TWEE KEER beantwoord, sien slegs die EERSTE poging na.
- As 'n kandidaat 'n antwoord van 'n vraag doodtrek (kanselleer) en nie oordoen nie, sien die doodgetrekte (gekanselleerde) poging na.
- Volgehoue akkuraatheid (CA) word in ALLE aspekte van die nasienriglyne toegepas; dit hou egter op by die tweede berekeningsfout.
- Wanneer 'n kandidaat aflesings vanaf 'n grafiek,, tabel, uitlegplan en kaart aanbied en ekstra antwoorde gee, penaliseer vir elke ekstra item.
- Afronding tel as 'n afsonderlike punt.
- Die algemene beginsel van nasien is, as 'n leerder een fout maak, word een punt afgetrek.
- 'n Gevolgtrekkingspunt kan slegs gegee word indien $\frac{1}{3}$ van die totale punte vir die subvraag toegeken is.
- Geen penalisering vir afronding (NPR) nie as die eerste desimaal korrek is.

QUESTION/VRAAG 1 [26 MARKS/26 PUNTE] ANSWER ONLY FULL MARKS			
Q/V	Solution/Oplossing	Explanation/Verduideliking	T&L
* 1.1.1	D ✓✓ A	2A correct option (2)	M L1 E
* 1.1.2	E ✓✓ A	2A correct option (2)	MP L1 E
* 1.1.3	C ✓✓ A	2A correct option (2)	M L1 E
* 1.1.4	G ✓✓ A	2A correct option (2)	M L1 E
1.2.1	✓ MA 220 mm ÷ 1 000 = 0,22 m ✓ A	1MA ÷ 1 000 1A conversion (2)	M L1 E
* 1.2.2	A ✓✓ A	2A correct option. (2)	M L1 M
* 1.2.3	Number of bricks / Aantal stene ✓RT = 2 860 mm ÷ 220 mm = 13 ✓A	1RT correct values 1MA dividing 1A number of bricks (3)	M L1 M

Q/V	Solution/Oplissing	Explanation/Verduideliking	T&L
* 1.3.1	18 ✓✓A	2A correct number (2)	MP L1 E
* 1.3.2	Number of cross pieces / <i>Getal dwarsstutte</i> ✓RT $= 6 \times 3$ ✓A $= 18$ ✓A	1RT 6 1A multiply by 3 1A pieces (3)	MP L1 M
* 1.3.3	Chair support / <i>Rugleuningbalk</i> ✓✓RT	2RT correct option (2)	MP L1 E
1.3.4	1,9 cm ✓✓A	2A correct dimension NPU (2)	MP L1 E
1.3.5	Space between cross pieces: / <i>Opening tussen dwarsstutte:</i> ✓ RT $\text{Space/Opening} = 1,27 \times 10 \text{ mm}$ $= 12,7 \text{ mm}$ ✓ MCA	1RT correct value 1MCA simplification NPU (2)	MP L1 E
		[26]	

QUESTION 2 [31 MARKS/31 PUNTE]			
Q/V	Solution/Oplossing	Explanation/Verduideliking	T&L
2.1.1	View from the top. ✓✓ A Aansig van bo. OR/OF View of the landscape from a certain height above ground like from a satellite or drone. ✓✓ A Aansig van die landskap vanaf 'n seker hoogte bo die grond soos vanaf 'n satelliet of hommeltuig. OR/OF View from an elevated height. ✓✓ A Aansig vanaf 'n verhewe hoogte. OR/OF Birds-eye view. ✓✓ A Voëlperspektief.	2A correct explanation (2)	MP L1 E
2.1.2	Number of campers/Getal kampeerders ✓RT ✓MCA = 6 + 15 + 4 + 15 + 4 + 4 + 5 + 15 + 8 + 10 + 6 + 6 + 20 = 118 ✓CA OR/OF Number of campers/Getal kampeerders = 20 + 15(3) + 10 + 8 + 6(3) + 5 + 4(3) ✓RT ✓MCA = 118 ✓CA	1RT all correct values 1MCA adding values 1CA simplification AO OR/OF 1RT all correct values 1MCA adding values 1CA simplification AO (3)	MP L1 E
2.1.3	South West/SW/Suidwes SW ✓✓ A	2A compass direction (2)	MP L1 E
* 2.1.4	2 ✓✓ O	2O identifying correct site (2)	MP L2 M
2.1.5	$\frac{3}{13} \times 100\%$ ✓A ✓A = 23,076923% ✓CA	1A correct numerator 1A correct denominator 1CA simplification NPR (3)	P L2 D
2.1.6 (a)	9 showers/storte ✓✓ A	2A correct number (2)	MP L1 E

Q/V	Solution/Oplossing	Explanation/Verduideliking	T&L
* 2.1.6 (b)	$\checkmark A$ $90 \text{ mm} = 8,2 \text{ m}$ $\checkmark C$ $90 \text{ mm} = 8\,200 \text{ mm} (\div 90)$ $\checkmark M$ $1 : 91,11$ $1 : 91$ $\checkmark R$ OR/OF $\checkmark A$ $9 \text{ cm} : 8,2 \text{ m}$ $9 \text{ cm} : 820 \text{ cm}$ $\checkmark C$ $= 1 : 91,11..$ $\checkmark M$ $= 1 : 91$ $\checkmark R$ Accept $\pm 1 \text{ mm}$ deviation on measurement per province Aanvaar $\pm 1 \text{ mm}$-afwyking op meting per provinsie	 1A measured distance 1C convert to mm 1M divide by 90 1R rounded answer OR/OF 1A measured distance 1C convert to cm 1M divide by 9 1R rounded answer (4)	MP L2 M
2.1.6 (c)	To allow people to sit while waiting for a toilet or shower to become available. $\checkmark \checkmark O$ <i>Vir mense om op te sit terwyl hulle wag dat 'n toilet of stort beskikbaar word.</i> OR/OF To place your clothes or belongings on while you are showering. <i>Om jou klere of besittings neer te sit terwyl jy stort.</i> OR/OF To sit while you change your outfit, or getting dressed or apply body lotion or for baby nappy change. <i>Om op te sit terwyl jy jou uitrusting verander of terwyl jy aantrek of lyfroom aansmeer of babadoeke verander.</i>	 2O reason (2)	MP L4 M
2.2.1	Day 2 / Dag 2 $\checkmark \checkmark A$	2A correct description (2)	MP L2 E
* 2.2.2	D $\checkmark \checkmark RT$	2RT correct option (2)	MP L2 M
* 2.2.3	12,5 km $\checkmark \checkmark RT$	2RT correct distance (2)	MP L2 E

Q/V	Solution/Oplossing	Explanation/Verduideliking	T&L
2.2.4	The part shows a continuous downward slope, it is downhill. <i>Die part het 'n aaneenlopende afwaartse helling getoon, dit is afdraand.</i> ✓✓O OR/OF That part does not have many uphill. <i>Daardie deel het nie baie opdraandes nie.</i>	20 correct explanation (2)	MP L4 E
* 2.2.5	Difference in height/<i>Verskil in hoogte</i> ✓RT ✓RT Difference/<i>Verskil</i> = 1 050 m – 900 m = 150 m ✓O He is CORRECT. / <i>Hy is KORREK</i>	1RT 1st correct value 1RT 2nd correct value 10 conclusion (3)	MP L4 M
			[31]

QUESTION/VRAAG 3 [31 MARKS/PUNTE]

Q/V	Solution/Oplossing	Explanation/Verduideliking	T&L
* 3.1.1	$\checkmark$ MA $= 14:13 - 12:55$ $\checkmark$ A $= 1 \text{ hour } 18 \text{ minutes}/1 \text{ uur } 18 \text{ minute}$ OR/OF 78 minutes OR/OF 1,3 hours/uur	1MA subtracting time 1A simplification [1hr18min] AO (2)	M L2 M
* 3.1.2	Total height of 4 pillows/ <i>Totale hoogte van 4 kussings</i> $= 11 \text{ cm} \times 4$ $\checkmark$ MA $= 44 \text{ cm}$ $\checkmark$ CA Difference/ <i>Verskil</i> $\checkmark$ RT $= 48 \text{ cm} - 44 \text{ cm}$ $= 4 \text{ cm}$ $\checkmark$ CA OR/OF $\checkmark$ RT $\checkmark$ MA $\checkmark$ MA Difference = $48 \text{ cm} - 11 \text{ cm} - 11 \text{ cm} - 11 \text{ cm} - 11 \text{ cm}$ $= 4 \text{ cm}$ $\checkmark$ CA	1MA multiplying by 4 1CA simplification 1RT height 1CA simplification OR/OF 1RT height 1MA subtracting 11 cm 1MA subtracting all the 11's 1CA simplification AO (4)	M L2 E
3.1.3	Perimeter = 2 (length + width)/ <i>Omtrek = 2 (lengte + breedte)</i> Perimeter/ <i>Omtrek</i> = 2 (46 cm + 30 cm) $\checkmark$ SF $= 2 (76 \text{ cm})$ $= 152 \text{ cm}$ $\checkmark$ CA Total length for 4 bags/ <i>Totale lengte vir 4 sakke</i> $= 152 \times 4$ $\checkmark$ MA $= 608 \text{ cm}$ $= \frac{608 \text{ cm}}{100}$ $= 6,08 \text{ m}$ $\checkmark$ C $\checkmark$ R $\therefore$ she must buy 6,5 m/ <i>Sy moet 6,5 m koop</i> OR/OF	1SF correct substitution 1CA simplification 1MA multiply by 4 1C simplification 1R correct rounding OR/OF	M L3 M

Q/V	Solution/Oplossing	Explanation/Verduideliking	T&L
	$30 \text{ cm} \div 100 = 0,3 \text{ m}$ $46 \text{ cm} \div 100 = 0,46 \text{ m}$ ✓C Perimeter/Omtrek = $2(0,3 \text{ m} + 0,46 \text{ m})$ ✓SF $= 1,52 \text{ m}$ ✓CA Total / Totaal = $1,52 \text{ m} \times 4$ ✓MCA $= 6,08 \text{ m}$ $\therefore$ she must buy 6,5 m/ <i>Sy moet 6,5 m koop</i> OR/OF Using ½ metre lengths/ Gebruik ½ metre lengtes Perimeter/Omtrek = $2(46 \text{ cm} + 30 \text{ cm})$ ✓SF $= 152 \text{ cm}$ ✓CA Total length for 4 bags / Totale lengte vir 4 sakke $= 152 \times 4$ ✓MA $= 608 \text{ cm}$ $\frac{1}{2} \text{ m} = 50 \text{ cm}$ ✓C Number of half metre lengths / Getal half-meter lengtes $= 608 \text{ cm} \div 50 \text{ cm}$ $= 12,16$ ≈ 13 ✓R	1C metre 1SF correct substitution 1CA simplification 1MCA multiply by 4 1R correct rounding OR/OF 1SF correct substitution 1CA simplification 1MA multiply by 4 1C to centimetre 1R correct rounding (5)	
* 3.2.1	Circumference / Omtrek $= 3,142 \times 8 \text{ cm}$ ✓SF $= 25,136 \text{ cm}$ ✓A	1SF substitute diameter 1A simplification NPR AO (2)	M L2 E
3.2.2	Radius/Radius $= \frac{8 \text{ cm}}{2}$ ✓MA $= 4 \text{ cm}$ ✓A	1MA concept of radius 1A simplification NPU AO (2)	M L1 E

Q/V	Solution/Oplossing	Explanation/Verduideliking	T&L
3.2.3	Area of a circle/Area/Oppervlakte van 'n sirkel $\checkmark \text{SF}$ $= 3,142 \times (4 \text{ cm})^2$ $= 3,142 \times 16 \text{ cm}^2 \quad \checkmark \text{MCA}$ $= 50,272 \text{ cm}^2 \quad \checkmark \text{CA}$ Area to be painted / Oppervlakte wat geverf moet word $\text{Area} = 50,272 - 0,3142 = 49,9578 \text{ cm}^2 \quad \checkmark \text{CA}$ $\text{Total area / Totale opp.} = 36 \times 49,9578 \text{ cm}^2$ $= 1\,798,4808 \text{ cm}^2. \quad \checkmark \text{MCA}$ $\text{Total area in m}^2 / \text{Totale opp. in m}^2$ $= 1\,798,4808 \div 100^2$ $= 0,179848 \text{ m}^2. \quad \checkmark \text{C}$ $6 \text{ m}^2 = 1 \ell = 1000 \text{ ml}$ $\checkmark \text{MA}$ $\dots \text{ m}^2 = 50 \text{ ml}$ $0,3 \text{ m}^2 = 50 \text{ ml}$ $0,3 \text{ m}^2 > 0,179848 \text{ m}^2 \quad \checkmark \text{CA}$ Therefore 50 ml will be more than sufficient. / $\checkmark \text{O}$ <i>Daarom sal 50 ml meer as genoeg wees.</i> OR/OF Area of a circle/Area/Oppervlakte van 'n sirkel $\checkmark \text{SF}$ $= 3,142 \times (4 \text{ cm})^2 \quad \checkmark \text{MCA}$ $= 50,272 \text{ cm}^2 \quad \checkmark \text{CA}$ Area to be painted/Oppervlakte wat geverf moet word $\text{Area} = 50,272 \text{ cm}^2 - 0,3142 \text{ cm}^2 = 49,9578 \text{ cm}^2 \quad \checkmark \text{CA}$ $\text{Total area / Totale oppervlakte}$ $= 36 \times 49,9578 \text{ cm}^2 = 1\,798,4808 \text{ cm}^2. \quad \checkmark \text{MCA}$ $= 1\,798,4808 \div 100^2$ $= 0,179848 \text{ m}^2 \quad \checkmark \text{C}$	CA from Question 3.2.2 1SF correct substitution 1MCA squaring 1CA simplification 1CA difference 1MCA multiply by 36 1C dividing by 10 000 or 100^2 1MA using ratio 1CA comparing areas. 1O verification OR/OF 1SF correct substitution 1MCA squaring 1CA simplification 1CA difference 1MCA multiply by 36 1C dividing by 10 000 or 100^2	M L4 M

Q/V	Solution/Oplossing	Explanation/Verduideliking	T&L
	Paint/ verf 6 m^2 with/met 1ℓ $\therefore 0,17984808 \text{ m}^2$ with /met $n \ell$ $n = \frac{0,17984808}{6} \ell$ ✓MCA $= 0,02997466 \ell$ $\approx 30 \text{ m}\ell$ ✓CA VALID / GELDIG ✓O OR/OF Area of ONE circle/Oppervlakte van EEN sirkel $\checkmark \text{SF}$ $= 3,142 \times (4 \text{ cm})^2$ $\checkmark \text{MCA}$ $= 50,272 \text{ cm}^2$ ✓CA Area to be painted/Oppervlakte wat geverf moet word $= 50,272 \text{ cm}^2 - 0,3142 \text{ cm}^2 = 49,9578 \text{ cm}^2$ ✓CA $6 \text{ m}^2 : 1 \ell$ $60\,000 \text{ cm}^2 : 1\,000 \text{ m}\ell$ ✓C $49,9578 : ?$ Paint needed/ Verf benodig $\frac{49,9578 \times 1\,000}{60\,000}$ ✓MCA $= 0,83263 \text{ m}\ell$ Paint for 36/ Verf vir 36 $= 0,83263 \text{ m}\ell \times 36$ ✓MCA $= 29,97 \text{ m}\ell$ ✓CA $\therefore$ VALID / GELDIG ✓O OR/OF Radius $= \frac{4 \text{ cm}}{100} = 0,04 \text{ m}$ ✓SF Area of circle/ Opp van sirkel $= 3,142 \times (0,04)^2$ ✓MCA $= 0,0050272 \text{ m}^2$ ✓CA Area of circular hole/ Opp van gaatjie $= \frac{0,3142}{10\,000}$ ✓C $= 0,0000314 \text{ m}^2$ Area to be painted $= 0,0050272 \text{ m}^2 - 0,00003142 \text{ m}^2$ $= 0,00499578 \text{ m}^2$ ✓CA $\therefore 0,00499578 \text{ m}^2 \times 36$ $= 0,17984808 \text{ m}^2$ ✓MCA Amount of paint/Hoeveelheid verf $= \frac{0,17984808}{6 \ell} \times 1\,000 \text{ m}\ell$ ✓MCA $= 29,97468 \text{ m}\ell \approx 30 \text{ m}\ell$ ✓CA $\therefore 30 \text{ m}\ell$ is less than $50 \text{ m}\ell$ VALID / GELDIG ✓O	1MCA using ratio 1CA paint needed 1O verification OR/OF 1SF correct substitution 1MCA squaring 1CA simplification 1CA difference 1C converting , 1MCA using ratio 1MCA multiply by 36 1CA paint needed 1O verification OR/OF 1SF correct substitution 1MCA squaring 1CA simplification 1C dividing by 10 000 1CA difference 1MCA multiply by 36 1MCA using ratio 1CA paint needed 1O verification	

Q/V	Solution/Oplossing	Explanation/Verduideliking	T&L
	OR/OF Area of a circle / <i>Oppervlakte van 'n sirkel</i> $= 3,142 \times 4^2$ ✓SF $= 3,142 \times 16$ ✓MCA $= 50,272 \text{ cm}^2$ ✓CA Area to be painted / <i>Oppervlakte wat geverf moet word</i> Area / <i>Opp</i> = $50,272 - 0,3142$ $= 49,9578 \text{ cm}^2$ ✓CA $6 \text{ m}^2 / \ell = 60\,000 \text{ cm}^2 / \ell$ $= 60\,000 \text{ cm}^2 / 1\,000 \text{ m}\ell$ ✓C Amount of paint for one tag / <i>Hoeveelheid van verf per houtplaatjie</i> $= 49,9578 \div 60\,000 \times 1\,000$ ✓MCA $= 0,83263 \text{ m}\ell$ Paint for 36 tags/ <i>Verf vir 36 houtplaatjies</i> $0,83263 \text{ m}\ell \times 36$ ✓MCA $= 29,97468 \text{ m}\ell$ ✓CA VALID / <i>GELDIG</i> ✓O	OR/OF 1SF correct substitution 1MCA squaring 1CA simplification 1CA difference 1C conversion 1MCA using ratio 1MCA multiply by 36 1CA paint needed 1O verification (9)	
* 3.3.1	Volume of a cube = side $\times$ side $\times$ side/ <i>Volume van 'n kubus</i> = $sy \times sy \times sy$ ✓SF ✓SF $2\,744 \text{ cm}^3 = \text{side} \times \text{side} \times \text{side}$ $(\text{side})^3 = 2\,744 \text{ cm}^3$ ✓MA $14 \times 14 \times 14 = 2\,744$ Side/Sy = 14 cm ✓CA	1SF substitution number 1SF cube unit 1 MA change subject of the formula 1CA simplification (4)	M L3 M
3.3.2	$8 + 7 = 15$ $P = \frac{15}{35}$ ✓A $= 0,42857...$ $\approx 0,43$ ✓CA OR/OF $P = \frac{8}{35} + \frac{7}{35}$ ✓A $= 0,22857... + 0,2$ ✓A $= 0,42857...$ $\approx 0,43$ ✓CA	1A numerator 1A denominator 1CA simplification OR/OF 1A denominator 1A writing as decimals 1CA simplification NPR (3)	P L2 E
		[31]	

QUESTION 4 [29 MARKS]			
Q/V	Solution/Oplissing	Explanation/Verduideliking	T/L
* 4.1.1	Kerosene or lamp oil /Keroseen of Lampolie ✓✓ RT	2RT correct product (2)	MP L1 E
4.1.2	Gasoline or petrol/ Brandstof of Petrol ✓✓ RT	2RT correct product (2)	MP L1 E
* 4.1.3	$^{\circ}\text{C} = \frac{(^{\circ}\text{F} - 32)}{1,8}$ $300^{\circ}\text{C} = \frac{(^{\circ}\text{F} - 32)}{1,8}$ $^{\circ}\text{F} = 1,8 \times 300 + 32$ $= 572$	1RT correct value 300 1SF substituting information correctly 1S changing subject of the formula 1CA simplification AO (4)	M L3 M
* 4.1.4	Surface area of an open cylinder/ Buite-oppervlakte van 'n oop silinder $= 3,142 \times \text{diameter} \times \text{height}$ $= 3,142 \times \text{deursnee} \times \text{hoogte}$ $= 3,142 \times 6 \text{ m} \times 54 \text{ m}$ $= 1\,018,008 \text{ m}^2$ Area of pipes/Oppervlakte van pype $= \frac{2,5}{100} \times \frac{1\,018,008}{1}$ $= 25,4502 \text{ m}^2$ Total Surface Area/Totale buiteoppervlakte $= 1\,018,008 \text{ m}^2 - 25,4502 \text{ m}^2 + 150,816 \text{ m}^2$ $= 1\,143,3738 \text{ m}^2$	1SF substitution 1CA simplification 1MCA percentage calculation 1CA simplification 1MCA subtracting pipe area 1MA adding A + C 1CA total surface area OR/OF	M L3 D

Q/V	Solution/Oplossing	Explanation/Verduideliking	T/L
	Surface area of an open cylinder/<i>Buite-oppervlakte van 'n oop silinder</i> SA = $3,142 \times \text{diameter} \times \text{height}$/ BO = $3,142 \times \text{deursnee} \times \text{hoogte}$ = $3,142 \times 6 \text{ m} \times 54 \text{ m}$ ✓ SF = $1\,018,008 \text{ m}^2$ ✓ CA Excluding area of pipes/<i>Oppervlakte van pype uitgesluit</i> ✓ MA ✓ MCA Area (excluding)/<i>Opp(uitgesluit)</i> = $\frac{97,5}{100} \times \frac{1\,018,008}{1}$ = $992,5578$ ✓ CA Total SA/<i>Totale BO</i> = $992,5578 \text{ m}^2 + 150,816 \text{ m}^2$ ✓ MA = $1\,143,3738 \text{ m}^2$. ✓ CA	1SF substitution 1CA simplification 1MA less 2,5% 1MCA percentage calculation 1CA simplification 1MA adding A + C 1CA total surface area NPR (7)	
* 4.2.1	Number of bricks in 1 row of a double brick wall <i>Getal stene in een ry van 'n dubbelsteenmuur</i> = 19 ✓✓ RT Number of bricks for 1 garage door <i>Getal stene vir 1 motorhuis deur</i> = 19×20 ✓ A = 380 Total number of bricks needed /<i>Totale getal stene nodig</i> = 380×2 ✓MCA = 760 ✓R ∴ 2 Pallets of bricks /<i>Stapelborde met stene</i> OR/OF	2RT bricks in double row 1A number of layers 1MCA doubling 1R number of pallets OR/OF	M L2 M

Q/V	Solution/Oplossing	Explanation/Verduideliking	T/L
	$\begin{aligned} & \checkmark \text{ RT } \checkmark \text{ RT} \\ \text{Single line wall/Enkelmuur} &= 20 \times 10 \\ &= 200 \text{ bricks/ stene} \\ & \checkmark \text{ CA} \\ \text{Double line wall/Dubbelmuur} &= 2 \times 200 = 400 \text{ bricks} \\ \text{To cover space of two garage doors:/} \\ \text{Om die spasie van twee motorhuisdeure te dek} \\ \text{Number of bricks/Getal stene} \\ &= 2 \times 400 \checkmark \text{ MCA} \\ &= 800 \\ \text{Number of pallets needed/} \\ \text{Stapelborde met stene benodig} \\ &= 2 \checkmark \text{ R} \end{aligned}$	1RT bricks (height) 1RT bricks (row) 1CA bricks on double walls 1MCA doubling 1R number of pallets (5)	
4.2.2	$\begin{aligned} & \checkmark \text{ MA } \checkmark \text{ SF} \\ \text{Area of 2 doors/Opp van 2 deure} &= 2 \times 2,13 \times 3 \\ &= 12,78 \text{ m}^2 \checkmark \text{ A} \\ \text{Labour cost/Arbeidskoste} &= 12,78 \text{ m}^2 \times \text{R}500 \\ &= \text{R } 6\,390 \checkmark \text{ CA} \\ \text{Brick cost/Steenkoste} &= 2 \times 525 \times \text{R}6,45 \\ &= \text{R } 6\,772,50 \checkmark \text{ CA} \\ \text{COST} &= \text{Other material} + \text{Labour} + \text{Bricks cost/} \\ \text{KOSTE} &= \text{Ander materiaal} + \text{Arbeid} + \text{Steenkoste} \\ \text{Total cost/Totale koste} &= \text{R}2\,000 + \text{R}6\,390 + 6\,772,50 \\ &= \text{R}15\,162,50 \checkmark \text{ CA} \\ \text{Not VALID/Nie GELDIG.} & \checkmark \text{ O} \\ & \text{OR/OF} \end{aligned}$	CA pallets from 4.2.1 1MA doubling 1SF correct values 1A simplification 1CA labour cost 1CA brick cost 1CA amount 1O verification OR/OF	M/Fin L4 M

Q/V	Solution/Oplossing	Explanation/Verduideliking	T/L
	For one door/ Vir een deur Area/ oppervlakte = $2,13 \text{ m} \times 3 \text{ m}$ ✓ SF $= 6,39 \text{ m}^2$ ✓ A Labour cost/Arbeidskoste = $6,39 \text{ m}^2 \times \text{R}500$ $= \text{R}3\,195$ ✓ MCA ∴ for two / vir twee = $2 \times \text{R}3\,195$ $= \text{R}6\,390$ ✓ CA Cost per pallet/ koste per stapelbord = $\text{R}6,45 \times 525$ $= \text{R}3\,386,25$ ∴ for two / vir twee = $2 \times \text{R}3\,386,25$ $= \text{R}6\,772,50$ ✓ CA Total cost/Totale koste = $\text{R}2\,000 + \text{R}6\,390 + \text{R}6\,772,50$ $= \text{R}15\,162,50$ ✓ CA Not VALID/Nie GELDIG. ✓ O OR/OF ✓ SF Area of 2 doors / Opp van 2 deure = $2 \times 2,13 \times 3$ $= 12,78 \text{ m}^2$ ✓ MA $\approx 13 \text{ m}^2$ (cost per m^2) ✓ CA Labour Cost / Arbeidskoste = $13 \times \text{R}500$ $= \text{R}6\,500$ ✓ CA Brick cost / Steen koste = $2 \times 525 \times \text{R}6,45$ $= \text{R}6\,772,50$ ✓ CA Total Cost / Totale Koste = $\text{R}6\,500 + \text{R}6\,772,50 + \text{R}2\,000$ $= \text{R}15\,272,50$ ✓ CA Not VALID / Nie GELDIG nie ✓ O	1SF correct values 1A simplification 1MCA labour cost 1CA doubling 1CA brick cost 1CA amount 1O verification OR/OF 1SF substitution 1MA doubling 1CA simplification 1CA labour cost 1CA brick cost 1CA amount 1O verification	(7)
4.2.3	✓✓ O To be easily lifted by a fork lift./Om maklik met 'n vurkhyser op te lig word. OR/OF Storage/ loading / transportation are made easier /Berging/ laai / vervoer word vergemaklik. OR/OF It keeps them from breaking / Dit keer dat hulle breek OR/OF It makes counting easier/Dit is makliker om te tel To protect the bricks from damage/ Dit beskerm die stene van beskadiging To keep it tight and compact/ Om die stene stewig en kompak te hou	2O reason	M L4 E (2)
		[29]	

QUESTION 5 [33 MARKS]			
Q/V	Solution/Oplossing	Explanation/Verduideliking	T/L
5.1.1	✓RT Adelaide and/en Melbourne ✓RT	1RT 1 st city 1RT 2 nd city (2)	MP L1 E
* 5.1.2	✓RT ✓RT Aug, Sep, Oct, Nov ✓RT Aug, Sep, Okt, Nov	1RT first correct month 1RT second correct month 1RT ALL correct months (3)	MP L1 M
5.1.3	✓RT 14 655 – (738 + 1 062 + 922 + 1 705 + 2 850 + 1 871 + 2 811) = 14 655 – 11 959 ✓MCA = 2 696 km ✓CA	1RT all correct values 1MCA subtracting 1CA simplification AO (3)	MP L2 M
* 5.1.4	Tasmania./Tasmanië ✓✓RT	2RT correct state (2)	MP L1 E
5.1.5	Distance/Afstand = Speed × time 1 705 km = s × 20 h 40 min ✓SF 1 705 km = s × 20,67 h ✓C Speed/Spoed = $\frac{1\,705\text{ km}}{20,67\text{ h}}$ ✓S = 82,5 km/h ✓CA	1SF substitution 1C converting to hours 1S change subject of formula 1CA simplification (4)	M L2 M
5.2.1	✓RT 1 142 feet/voet = 348 m ✓RT $1\text{ foot/voet} = \frac{348}{1\,142}$ = 0,304728546 1 foot/voet ≈ 0,305 m ✓A	1RT 1 142 1RT 348 1A 0,305 (3)	M L2 M
* 5.2.2	Uluru : Eiffel Tower : Big Ben/ Uluru : Eiffeltoring : Big Ben 348 : 324 : 96 ✓RT ✓A = 29 : 27 : 8 ✓✓CA	1RT correct values 1A correct order 2CA simplified ratio (4)	M L2 E

Q/V	Solution/Oplissing	Explanation/Verduideliking	T/L
* 5.3.1	$P = 22,5\%$ ✓✓A OR/OF $P = \frac{1\,729\,742}{7\,688\,220}$ ✓✓A $= 0,224986... \text{ or } 22,5\%$	2A correct probability (2)	P L2 E
5.3.2	$\text{Area of island/Opp van eiland} = \frac{32\,159}{64\,519}$ ✓RT $= 0,498 \approx 0,5$ ✓CA OR/OF $\text{Half of Tasmania / Helfte van Tasmanië}$ $\frac{1}{2} \times 64\,519 = 32\,259,5 \text{ km}^2$ ✓MA $\text{Islands / Eilande} = 32\,159 \text{ km}^2$ ✓RT $\frac{64\,519}{32\,159} : \frac{32\,159}{32\,159}$ ✓RT $\approx 2 : 1$ ✓CA OR/OF	1RT area of islands 1RT area of Tasmania 1CA simplification OR/OF 1RT Tasmania area 1MA simplification 1RT island area OR/OF 1RT area of islands 1RT area of Tasmania 1CA simplification (3)	MP L4 M
* 5.3.3 (a)	$\text{Population density/Bevolkingsdigtheid}$ $= \frac{454\,499}{2\,358}$ ✓RT $= 192,74...$ ✓RT ≈ 193 ✓R	1RT correct 454 499 1RT correct area 2 358 1R simplification rounded up (3)	M L3 D
* 5.3.3 (b)	$\text{Land \% area/Land \% opp} = \frac{2\,358}{7\,688\,220} \times 100\%$ ✓RT $= 0,0306..%$ ✓RT $\text{Rounds off to zero/Rond af na nul.}$ ✓CA ≈ 0 ✓O	1RT correct 2 358 1RT 7 688 220 1CA simplification 1O opinion (4)	M L4 D
		[33]	